

Bapa Rao Pendyala

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Experience

Software Development Intern, Genset AI Solutions

May 2026 – July 2026

Smart Community Management System | Node.js, Express.js, MongoDB, React.js

- Built a full-stack Community Management System with **Node.js**, **Express.js**, **MongoDB**, and **React.js**, featuring **90+ REST APIs** with **JWT auth** and **RBAC**.
- Designed a responsive React UI with **CSS**, including a reusable component library and role-based dashboards for **5 user roles**.
- Enhanced security using **bcrypt hashing** and **JWT refresh tokens**, validated via **90+ automated API tests**.

Machine Learning Intern, Universiti Teknologi PETRONAS (UTP) [Certificate] [GitHub]

April 2025 – July 2025

Python, scikit-learn, Random Forest, SVM, SMOTE

Malaysia (Remote)

- Engineered a production-ready NLP classification pipeline for maritime incident reports, processing **3,000+ records** into 5 severity levels at **96%+ accuracy** using a Random Forest and SVM ensemble.
- Reduced class imbalance by **40%** with SMOTE and improved prediction reliability through probability calibration for risk-sensitive forecasts.
- Lowered false negatives by **25%** in safety-critical scenarios; the work earned a **3rd Place IEEE Best Paper Award** (EI 2025).

Projects

BeverageStore – Multi-Vendor E-Commerce Platform

MongoDB, Express.js, React.js, Node.js, Razorpay, Stripe, Socket.io

- Engineered a B2C and B2B marketplace with distinct vendor portals and granular access controls that reduced unauthorized data access by **70%**.
- Integrated Razorpay and Stripe payment gateways with a GST engine and flash-sale system, validating **50+ edge-case scenarios**.
- Built live order tracking that handled **100+ concurrent updates**, alongside an analytics dashboard covering revenue and conversion metrics.

Seva Portal – Government Service Platform [GitHub]

React.js, Node.js, Express.js, MySQL

- Built a full-stack civic portal serving several users, with **10+ endpoints** for digital form submission and live application tracking.
- Designed a normalized MySQL schema with referential integrity supporting concurrent multi-user document submissions.
- Implemented JWT authentication and multi-role RBAC across citizen, officer, and admin tiers for secure access control.

SyncStream – Real-Time Networked Folder Sync [GitHub]

C++17, TCP Sockets, Multithreading, POSIX, HTTP

- Built a bidirectional folder sync tool in C++17 over raw TCP sockets, syncing changes across networked machines in under **2 seconds**.
- Implemented a producer-consumer architecture with mutex-protected queues and condition variables, sustaining thread-safe concurrency across multiple workers.
- Developed an embedded HTTP dashboard for live monitoring and runtime configuration of the client-server system.

EDUCATION

SRM University, AP

Bachelor of Technology in CSE (AI & ML); CGPA: 9.5/10; Dean's List (2025–2026)

2023 – 2027

Rajiv Gandhi University of Knowledge Technologies (RGUKT), Nuzvid

Pre-University (Mathematics, Physics, Chemistry); CGPA: 9.96/10

2021 – 2023

Technical Skills

Programming Languages: C++, Java, Python, JavaScript, SQL

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC

Machine Learning: scikit-learn, Random Forest, SVM, SMOTE, TF-IDF, PyTorch

Databases & Tools: MongoDB, MySQL, Git, Linux (RHEL), Shell Scripting

Core CS: Data Structures, Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Soft Skills: Problem Solving, Team Leadership, Cross-functional Collaboration, Communication, Time Management, Adaptability

Achievements & Certifications

- ET-AI Hackathon 2026 Semi-Finalist** – Reached the semi-finals of The Economic Times AI Hackathon, a national-level competition (February 2026).
- GATE 2026 Qualified** – Computer Science and IT, conducted by IIT Guwahati among 211,020 candidates nationwide (February 2026).
- Red Hat Certified System Administrator (RHCSA)** – Linux administration, networking, storage, and shell scripting on RHEL (2025). [Credential]
- Oracle Certified Professional: Java SE 17 Developer** – OOP, streams, concurrency, and enterprise application development (February 2026). [Credential]